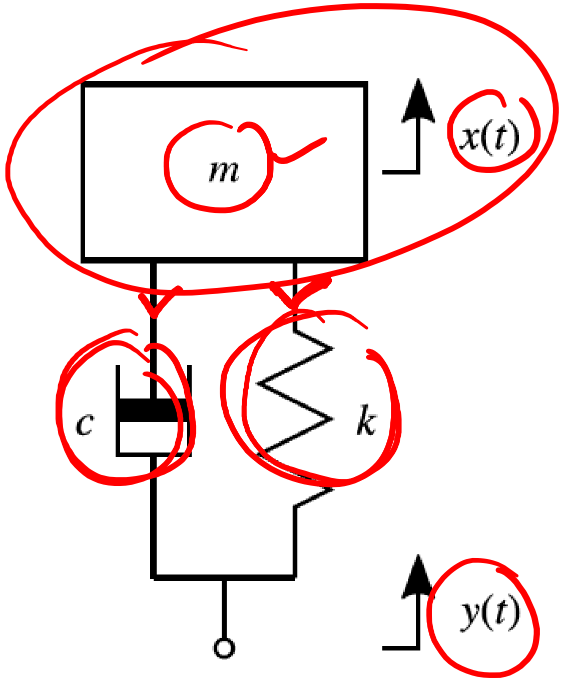
Example:
Block Diagram Representation of Control Systems



$$\uparrow \quad F = m\ddot{x}$$

$$-c(\dot{x}-\dot{y}) - k(x-y) = m\ddot{x}$$

$$-c\dot{x} + c\dot{y} - kx + ky = m\ddot{x}$$

$$c\dot{y} + ky = m\ddot{x} + c\dot{x} + kx$$

$$\mathcal{L} \Rightarrow \quad csY(s) + kY(s) = ms^2X(s) + csX(s) + kX(s)$$

$$Y(s)\{cs+k\} = X(s)\{ms^2+cs+k\}$$

$$T/f = \frac{X(s)}{Y(s)} = \frac{cs+k}{ms^2+cs+k}$$

$$Y(s) \longrightarrow \boxed{\frac{cs+K}{ms^2+cs+K}} \longrightarrow X(s)$$

1: Cascaded Blocks

$$X_1(s) \longrightarrow \boxed{G_1(s)} \xrightarrow{X_2(s)} \boxed{G_2(s)} \xrightarrow{X_3(s)}$$

$$\Longrightarrow \quad X_1(s) \longrightarrow \boxed{\;?\;} \longrightarrow X_3(s)$$

$$X_2(s) = X_1(s)\,G_1(s)$$

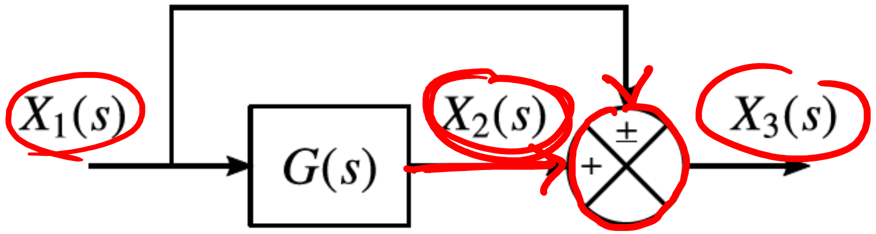
$$X_3(s) = \boxed{X_2(s)}\,\underline{G_2(s)}$$

$$X_3(s) = \big(X_1(s)\,G_1(s)\big)\,G_2(s)$$

$$X_3(s) = X_1(s)\big(G_1(s)\,G_2(s)\big)$$

$$X_1(s) \longrightarrow \boxed{G_1(s)\,G_2(s)} \longrightarrow X_3(s)$$

2: Summing Two Signals

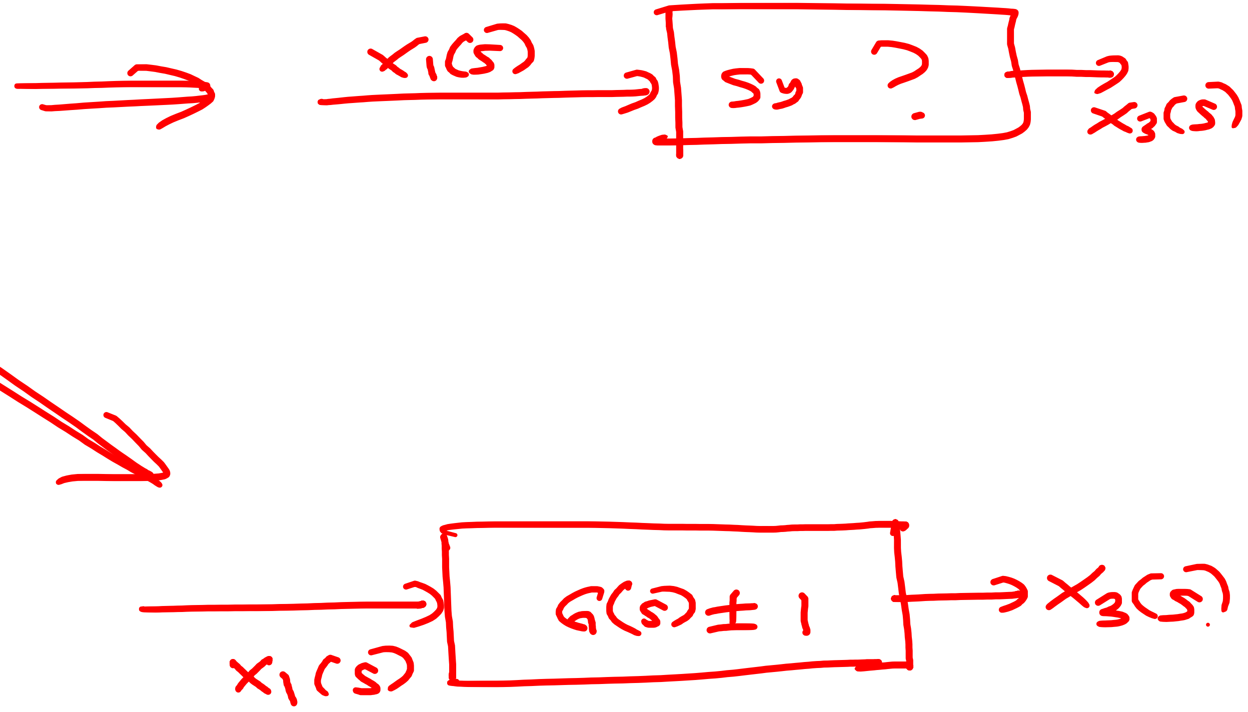


$$X_2(s) = X_1(s)G(s)$$

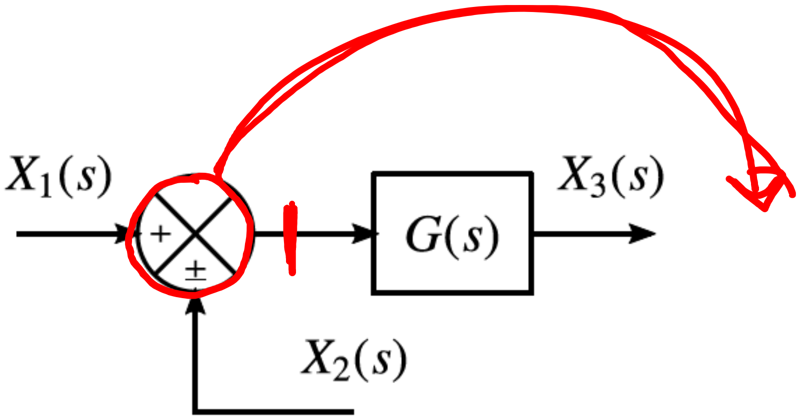
$$X_3(s) = X_2(s) \pm X_1(s)$$

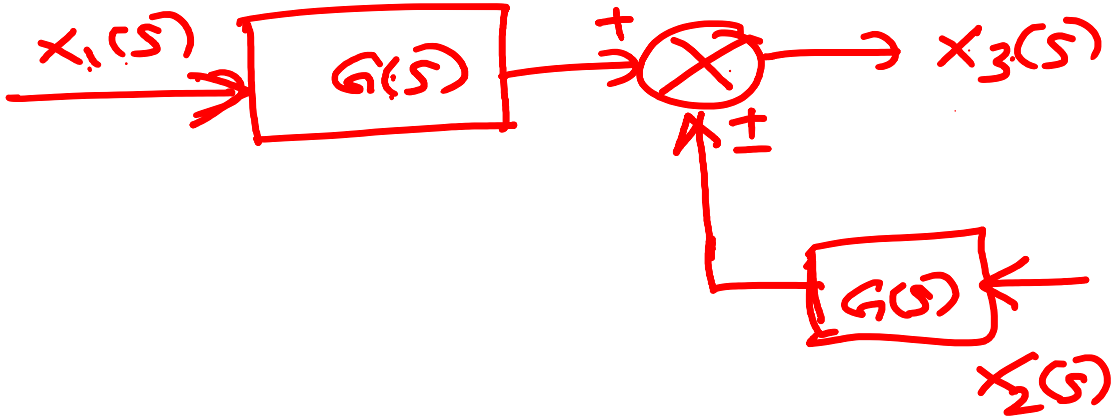
$$X_3(s) = X_1(s)G(s) \pm X_1(s)$$

$$X_3(s) = X_1(s)\underbrace{[G(s) \pm 1]}$$



3: Moving a Summing Point Behind a Block

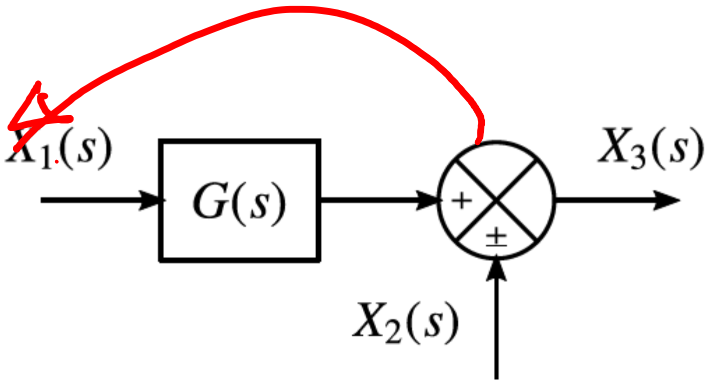


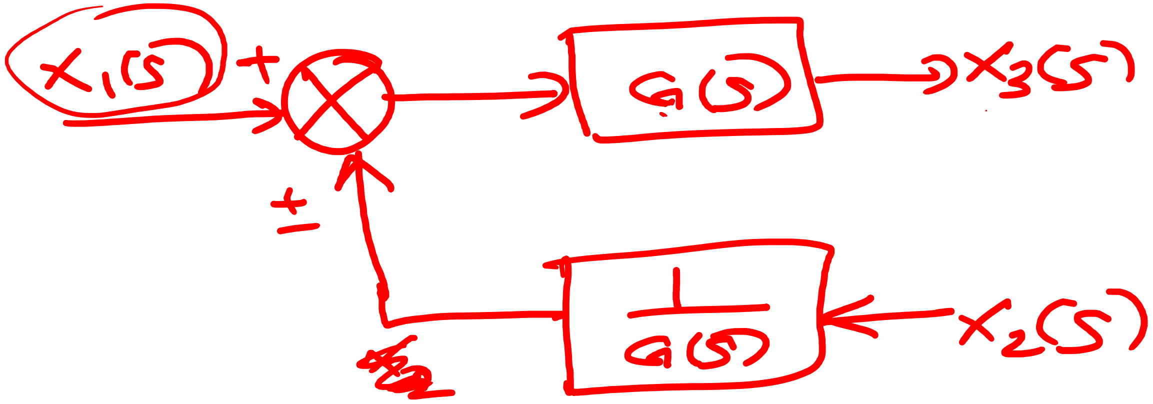


$$X_3(s) = [X_1(s) \pm X_2(s)]\, G(s)$$

$$X_3(s) = X_1(s)\, G(s) \pm X_2(s)\, G(s)$$

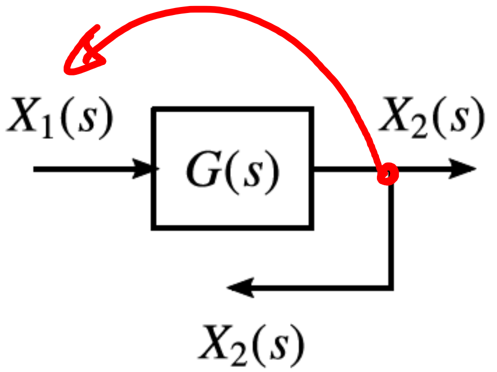
4: Moving a Summing Point Ahead of a Block



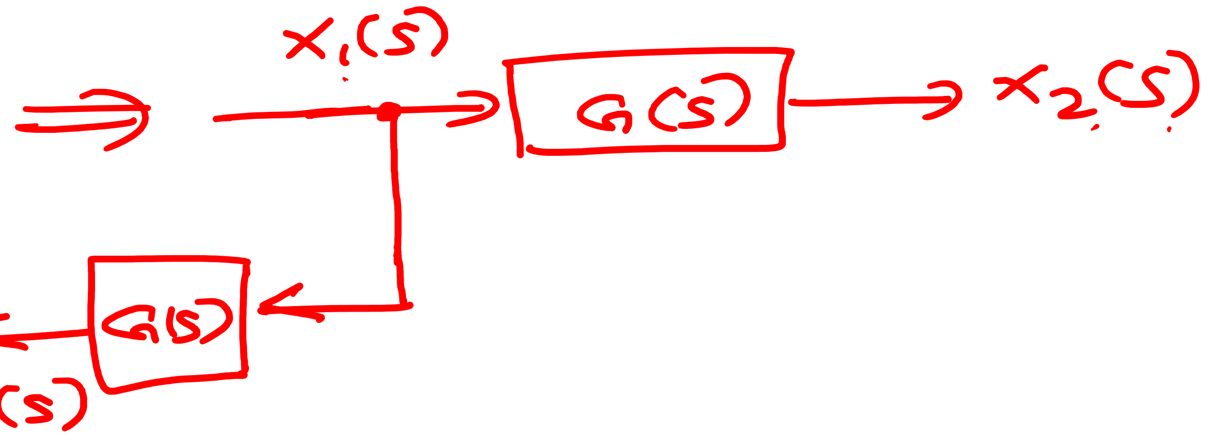


$$X_3(s) = X_1 G(s) \pm X_2(s)$$

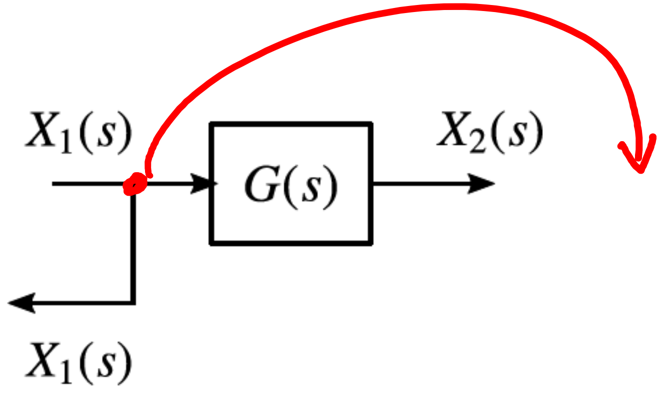
5: Moving a Branch Point Ahead of a Block



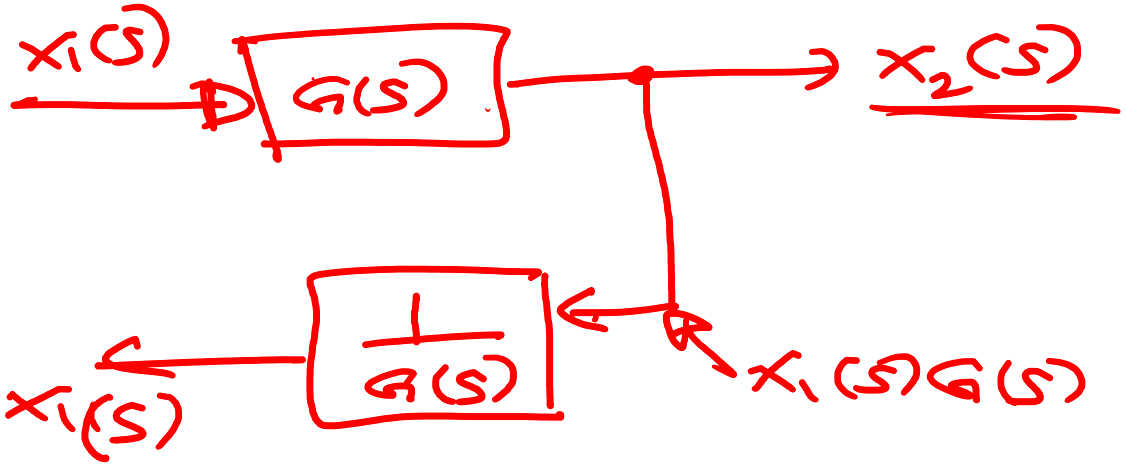
$$x_2(s) = x_1 G(s)$$



6: Moving a Branch Point Behind a Block

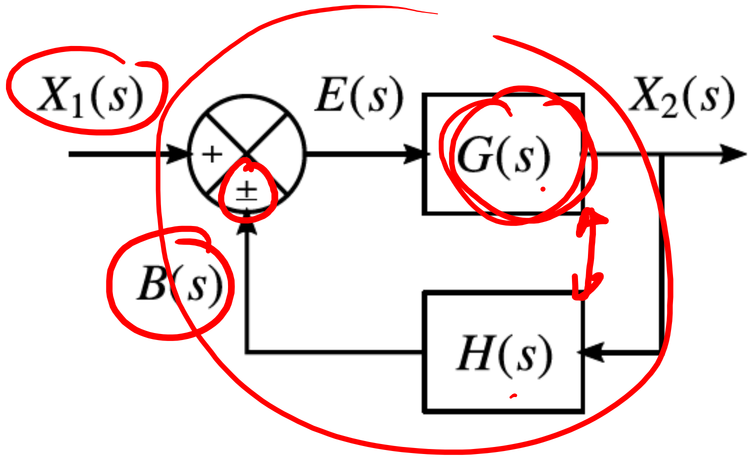


$\Rightarrow$

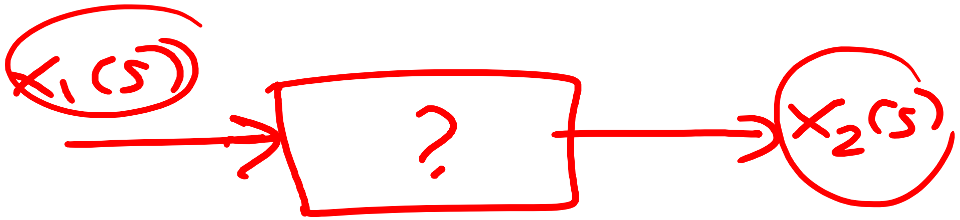


$$X_2(S) = X_1(S)\,G(S)$$

7: Eliminating a Feedback Loop



$\Longrightarrow$



$$B(s) = H(s)X_2(s)$$

$$E(s) = X_1(s) \pm B(s)$$

$$X_2(s) = E(s)G(s) = G(s)\left[X_1(s) \pm B(s)\right]$$

$$X_2(s) = G(s)\left[X_1(s) \pm H(s)X_2(s)\right] \Longrightarrow X_2(s)\left[1 \pm G(s)H(s)\right] = X_1 G(s)$$

$$\boxed{\frac{X_2(s)}{X_1(s)} = \frac{G(s)}{1 \pm G(s)H(s)}}$$